

Performance Analysis of Compact Size Slotted Microstrip Patch Antenna for X-band Applications

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Abstract— In this paper a compact slotted microstrip patch antenna has been designed and analyzed for X-band application from 8 to 12 GHz. The design was implemented in Rogers RT5880 substrate and optimized slot geometry for enhancing electromagnetic properties. The proposed antenna achieved a wideband impedance bandwidth more than 2000 MHz, good impedance matching and a relatively constant far field radiation pattern within the operating frequency range. In addition, optimal current distribution over small footprint (20x20mm²) also provides a higher radiation efficiency (>90%) and higher directional gain. Therefore, this new design can be used as a competitive candidate for a variety of compact and efficient X-band applications such as satellite communications, radar sensors and defense grade applications.

Keywords— Patch antenna, X-band, DGS, impedance bandwidth, radiation efficiency, far-field stability.

INTRODUCTION

As the amount of data on our contemporary communication networks increases and carrying of the expand in Internet of Things (IoT), microstrip patch antennas have been at the leading edge of creating wireless technologies that are smaller, lighter, and more effective. According to report of statistics, the number of connected devices is expected to exceed 28 billion by 2027 [2]. In addition, the amount of data being transmitted via cell phones globally is estimated to increase almost three times from 2024 to 2030 [1] creating a significant need for more efficient network capacity. Consequently, in order to fulfill the requirements of upcoming satellite and radar systems, microstrip patch size reduction must continue to progress along with efficiency and performance capabilities. Satellite and radar systems require high-efficiency, low-cost, easy-to-fabricate and small footprint antennas. Both industries are also growing rapidly; the satellite communication industry is projected to grow from USD 93.71 billion in 2024 to USD 210.35 billion by 2033 [3]; the X-band radar industry is projected to grow from USD 6.9 billion in 2020 to USD 10.0 billion by 2035 [4]. The X-band frequency range (8.0-12.0 GHz) is critical for many applications such as air traffic control, defense tracking and high-throughput space communications. One of the biggest challenges facing antenna designers today is to develop an

antenna design that achieves both high-gain and broad bandwidth operation in a very compact form factor.

The development of high-performance microstrip patch antennas for X-band and associated applications has been the main focus of recent research. Using a Rogers RT/duroid substrate, Sultana et al. in [5] constructed a dual-patch antenna operating at 7.8 GHz for WiMAX and X-band satellite communication, achieving remarkable performance with a return loss of -49.92 dB, a VSWR of 1.006, and a gain of 8.5 dB. Goud et al. demonstrated a peak gain of 8.2 dBi at 9.64 GHz with a 200 MHz bandwidth using a pentagonal patch antenna with a moon-shaped slot and a defective ground structure (DGS) [6]. For dual-band radar applications, Alagumurugesan et al. created a slotted rectangular patch antenna in [7] that operates in C-band (5.26 GHz, 7.62 GHz) and X-band (11.60 GHz), with gains ranging from 4.9 to 8.1 dB. A metamaterial-enhanced square patch antenna for Ku-band applications was studied by Vinoth and Vallikannu; it achieved -23 dB return loss at 19.1 GHz with 2 dB gain, which qualifies it for satellite broadcasting [8]. High efficiency and wide bandwidth for microwave transmission are highlighted in recent research on X-band patch antennas (8–12 GHz). An insert-fed patch antenna with a defective ground plane was built by Jeyapoornima et al. [9]. In modeling, it achieved a return loss of -34.03 dB, bandwidth of 2.1 GHz, and gain of 4.73 dB; in experimentation, it achieved -32.73 dB, 2.25 GHz, and 4.15 dB.

The primary challenge within X-band systems is the tradeoff between the performance of antennas and the size of antennas, limiting the development of small-scale radar/satellite hardware. In Section II, this work addresses the above issues and presents a DGS microstrip patch antenna that has a smaller footprint than other similar antennas [5-9] and analyzes geometrical optimization of proposed antenna. Section III presents performance characteristics of the proposed design simulated in CST microwave studio through FDTD solver in terms of some key performance indicators (such as gain, efficiency, and impedance bandwidth) against existing designs. Finally, compare this design with recent relevant literature.

DESIGN ANALYSIS

This proposed microstrip patch antenna utilizes Rogers RT5880 as its substrate ($\epsilon_r = 2.2$, $\tan\delta = 0.0009$) thickness of 0.79 mm. The microstrip line is used to feed the proposed antenna, thus allowing an efficient excitation of the radiating patch while retaining the advantages of a planar compact design. A $20 \times 20 \text{ mm}^2$ substrate supports a radiating patch of $12 \times 16 \text{ mm}^2$ while a 2.456 mm wide feed line is to ensure a good impedance matching at 50Ω . The progressive design evaluation and respective performance on each step (step-I to step-V) is shown in Fig. 1, and 2 respectively. Fig. 2 demonstrates the impact of structural modifications on impedance matching and resonant characteristics. Fig. 3 illustrates the front view and back view of the rectangular patch antenna with multiple optimized slots that are responsible for achieving multi-resonant characteristics. The optimized dimensions and slots for proposed design are listed in Table I and II. A partial ground plane of dimensions $7 \times 17 \text{ mm}^2$ is used to enhance the bandwidth and the radiation efficiency of the antenna. In order to obtain a stable resonant frequency the geometry of the patch has been significantly modified in the proposed antenna by introducing several slots: S1 (two symmetric vertical slots); S2 (two upper diagonal slots); S3 (two lower diagonal slots); the slots are located on the metallized ground plane, that consists of two rectangular sections separated by a wide horizontal slot (g2) and side gaps (g1), this defected parasitic ground plane in Fig. 3(b) emphasized the performance of the antenna that highlighted in step-V (Fig. 2).

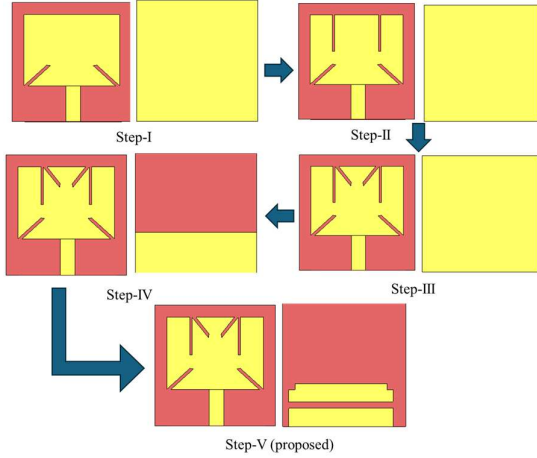


Fig. 1. Step-by-step design evaluation of the proposed antenna.

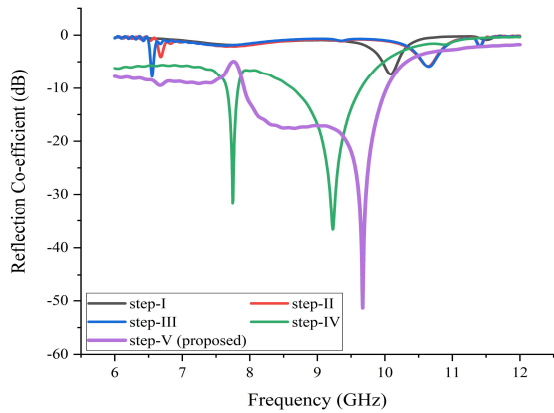


Fig. 2. Performance evaluation of the proposed antenna in terms of reflection coefficient.

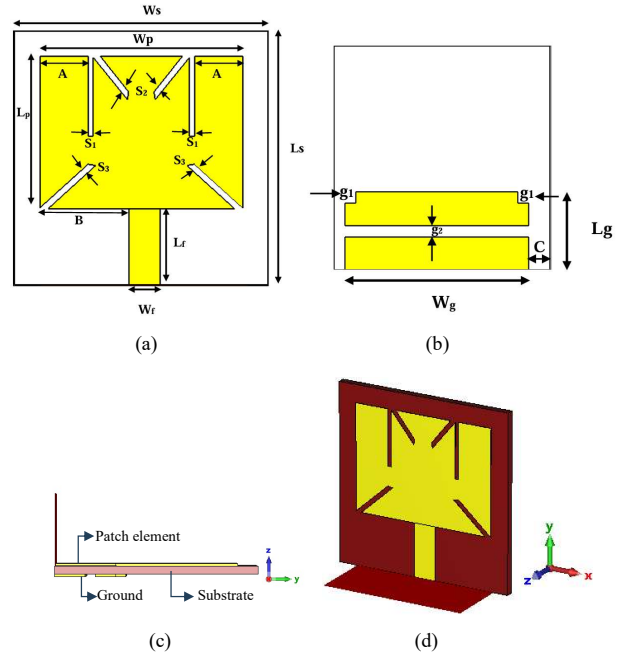


Fig. 3. (a) Front view, (b) back view, (c) side view, and (d) 3D view of the proposed antenna.

TABLE I. CALCULATE AND ROUND OFF THE ANTENNA DESIGN PARAMETERS

Antenna Parameters	Dimensional value (mm)	Antenna Parameters	Dimensional value (mm)
Substrate length, Width, $L_s = W_s$	20	Ground length, L_g	7
Patch Length, L_p	12	Ground width, W_g	17
Patch Width, W_p	16	Patch left arm, A	3.80
Feedline Length, L_f	6	Patch right arm, B	7
Feedline Width, W_f	2.456	Ground slot distance, C	2

TABLE II. SLOT PARAMETERS OF THE PROPOSED ANTENNA

Slot Parameters	Length (mm)	Width (mm)
S_1	6.30	0.40
S_2	4.29	0.61
S_3	5.12	0.61
g_1	1	1
g_2	1	17

PERFORMANCE ANALYSIS

In this section, the performance of the designed antenna is assessed through the evaluation in terms of return loss, impedance matching and far-field radiation characteristics of the proposed antenna model.

A. Reflection Co-efficient

Scattering parameter (S_{11}) represents the reflection coefficient that how much power is reflected back due to impedance mismatching between the feed line and the antenna at a given frequency. A smaller magnitude of S_{11} corresponds to a larger return loss, which signifies better impedance matching. Under this condition, analysis of S-parameter ($|S_{11}|$) for the proposed antenna presented in Fig. 4, demonstrates that the antenna has very good impedance matching at the resonant

frequency of 9.672 GHz with a minimum return loss of -51.33 dB. Therefore, the antenna has an effective impedance bandwidth of approximately 2.111 GHz which spans from 7.914 to 10.025 GHz and therefore meets the criteria for $S_{11} < -10$ dB. Furthermore, the deep resonance indicates that the antenna has a high radiation efficiency and small reflection loss, thus confirming a good energy transfer from the feed line to the radiating structure. The large return loss value further proves that the antenna is suitable for use in X-band applications and will provide a consistent level of performance and ensure reliable data transmission over the entire working frequency band.

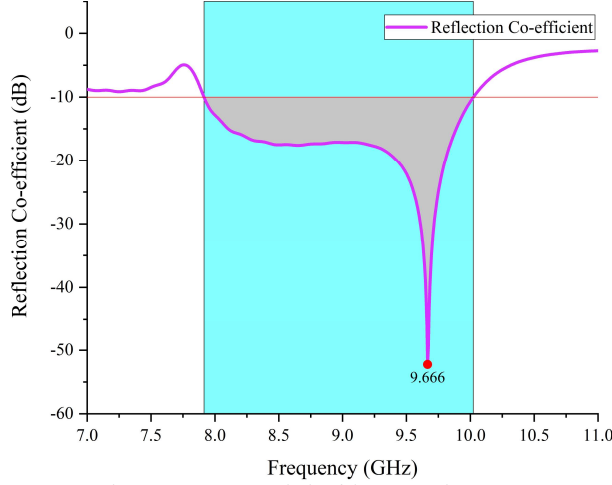


Fig. 4. S-parameter analysis of the proposed antenna.

B. Impedance Matching Performance Analysis

The simulated VSWR plot in Fig. 5 for the proposed antenna reveals a value of 1.006 at the resonant frequency. This result is exceptionally close to the ideal value of 1, indicating near-perfect impedance matching between an antenna and its feed line where all power from the feed line is delivered to the antenna and radiated. The real and imaginary components of the input impedance (Z_{11}) versus frequency are illustrated in Fig. 6. For a better result, the Z-matrix should be 50 ± 2 ohms for the real part and 0 ± 2 for the imaginary part. At the operating frequency of 9.67 GHz, the simulated input impedance is $Z_{in} = 48.59 + j0.44 \Omega$. This validates the strong resonance and minimal power reflection observed in the return loss plot.

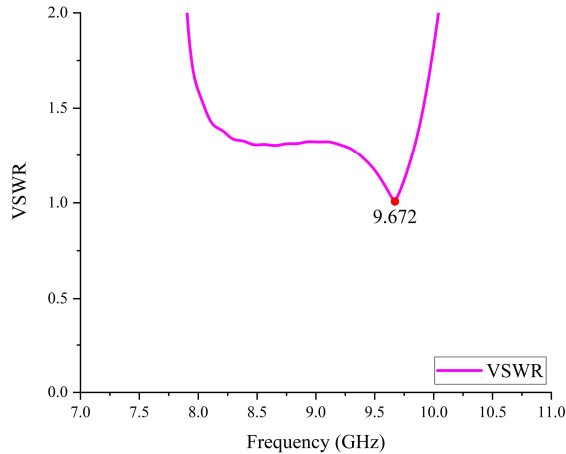


Fig. 5. VSWR analysis at resonance frequency of 9.672 GHz.

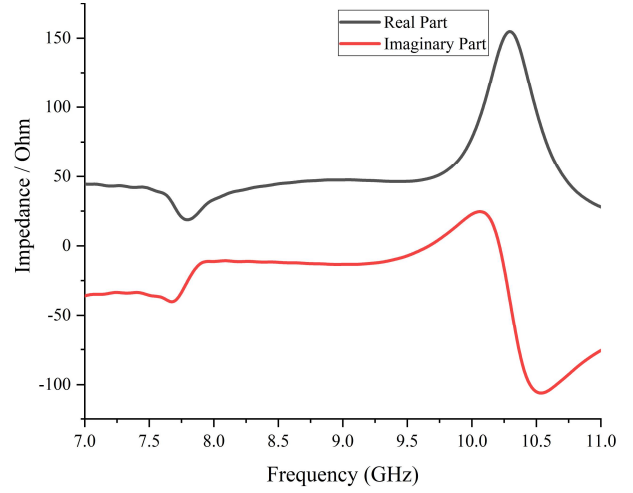


Fig. 6. Impedance (Z-matrix) analysis in real and imaginary parts.

C. Radiation Performance Analysis

The simulated antenna gain and directivity are presented in Fig. 7 as a function of frequency. At the central resonant frequency of 9.67 GHz, the antenna achieves a peak realized gain of 3.62 dB and a peak directivity of 3.82 dBi. According to Fig. 8, this corresponds to an estimated efficiency of 92.2% based on the following equation [10].

$$G = \eta \cdot D \quad (1)$$

Where, G = gain, η = radiation efficiency, D = directivity

The directivity of an antenna is the ratio of the maximum power density $P(\theta, \phi)_{max}$ to its average value over a sphere as observed in the far field of an antenna. The simulated 3D far-field directional characteristics and beam shape of the antenna are shown in Fig. 9(a) and 9(b), respectively.

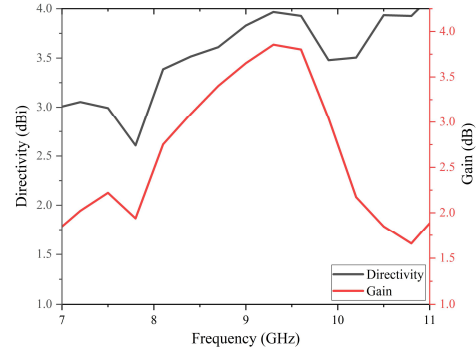


Fig. 7. Directivity and gain response of the proposed antenna.

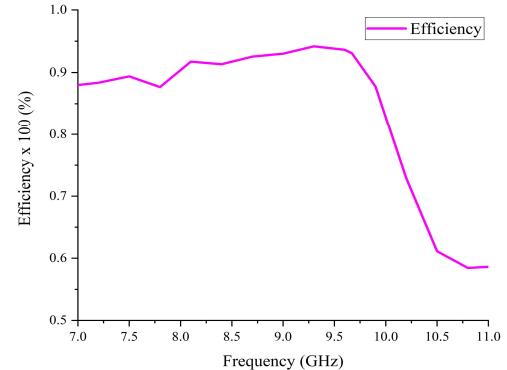


Fig. 8. Radiation efficiency of the proposed antenna.

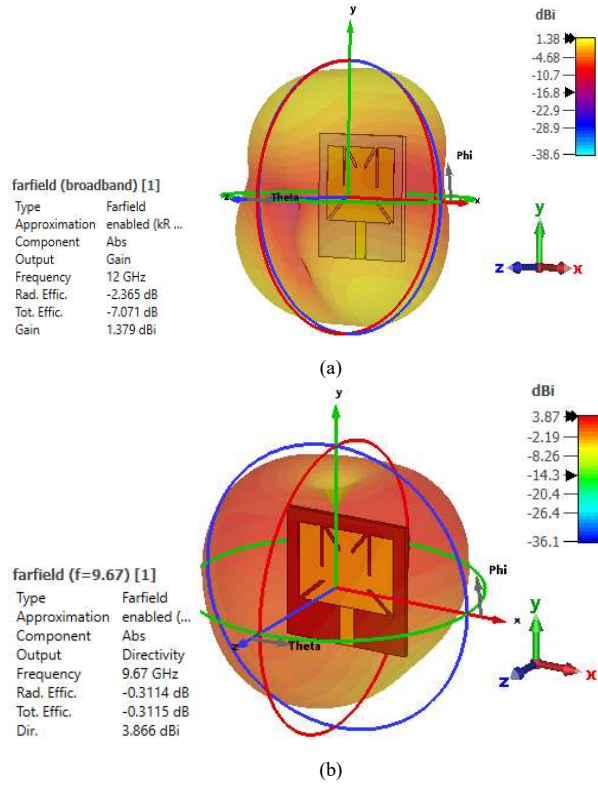


Fig. 9. 3D Radiation pattern with (a) gain and (b) directivity at 9.672 GHz.

The combined far-field radiation characteristics of the proposed antenna at 9.672 GHz in Fig. 10, illustrating both electric, magnetic field (E, H-field) distributions for $\phi = 0^\circ$ and $\phi = 90^\circ$. For the E-field patterns, the main lobe magnitudes around 18 dB(V/m) with directed at 1.0° and 22.0° corresponding angular widths of around 90° , respectively $\phi = 0^\circ, 90^\circ$. Similarly, at $\phi = 0^\circ, 90^\circ$ magnetic field main lobe directed in 1.0° and 22.0° in strength of -33 dB(A/m) with 3 dB beamwidths of 90° . In both plane side lobe exists strengthen of -1.9 dB and -1.1 dB respectively. The close similarity between E- and H-field distributions confirms the antenna's stable far-field response, symmetric radiation behavior, and consistent polarization performance at the resonant frequency of 9.672 GHz.

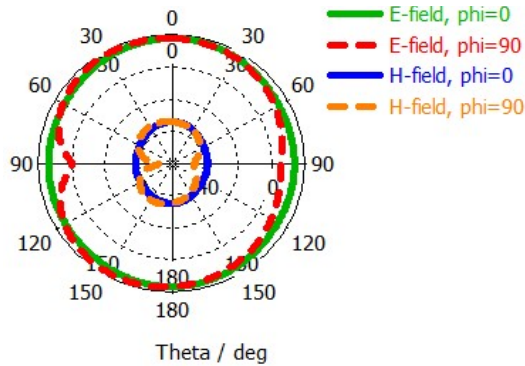


Fig. 10. Far-field characteristics of E, H-field at $\phi = 0^\circ, 90^\circ$ at resonance frequency of 9.672 GHz.

Fig. 11 depicts the simulated surface current density on both the radiating patch and the defective ground plane for the resonant frequency of 9.67 GHz. High current density is observed on the microstrip feed line as well as along the interior edges of the partially defective ground plane, further demonstrating the contribution of these areas to the operation of the antenna. The current flowing upward from the feed are primarily concentrated on the vertical edges of the radiating patch and in 'M'-shaped structure at the top and bottom diagonal slots (S_1, S_2, S_3 , and S_3) demonstrate how the edges are effectively disrupt the current flow, increasing the electrically effective length of the current flow and thereby controlling the resonant mode.

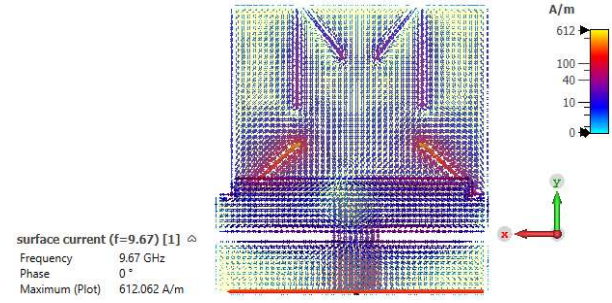


Fig. 11. Surface current distribution through the patch at the resonance frequency of 9.67 GHz.

COMPARATIVE ANALYSIS

Compared to the antennas in Table III, the proposed design (this work) exhibits the smallest size ($20 \times 20 \times 0.79 \text{ mm}^3$) and the widest bandwidth (2110 MHz), while maintaining excellent return loss (-51.33 dB) and the highest efficiency (92%). In contrast, earlier works report narrower bandwidths (160–900 MHz) and lower efficiencies ($<75.5\%$). Thus, the proposed antenna demonstrates clear improvement in miniaturization, impedance matching, and radiation efficiency for X-band applications.

TABLE III. COMPARISON WITH RECENT RELATED WORKS

Ref.	Size (mm ³)	Resonance Frequency (GHz)	BW (MHz)	Return loss (dB)	Max. Efficiency (%)
[5]	$80 \times 71.42 \times 71.42$	7.814	160	-49.92	75.5
[6]	$40 \times 40 \times 1.6$	9.55	250	-36.99	N/A
[7]	$24 \times 23.5 \times 2$	5.26, 7.62	900	-23.62	N/A
[8]	$40 \times 40 \times 1.6$	9.55	250	-17.17	N/A
[9]	$37 \times 47.6 \times 1.6$	10.152	2100	-34.03	N/A
<i>This work</i>	$20 \times 20 \times 0.79$	9.672	2110	-51.33	92

CONCLUSION

A compact and high-performance 'M' slotted microstrip patch antenna was developed for X-band operation. This slotted patch element used defected ground structure to achieve enhanced gain, radiation efficiency and impedance bandwidth. The wideband result upon this design confirms broadband operation for high data rate applications at 9.67 GHz operating frequency making effective performance of 92% high radiation efficiency, implying effective power radiation and minimal losses. Additionally, its stable E- and H-plane radiation patterns are observed with a realized gain of

3.62 dB make it a promising candidate for satellite, radar, and military communication systems. Future studies will focus on experimental validation, use array elements in different combinations, and create multifunctional antenna system.

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